

BLUESTOCK FINTECH

Mutual Fund Analytics Capstone

FINAL PROJECT REPORT

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Program	Data Analyst Internship — Cohort 2026
Division	Bluestock Fintech — Fintech Analytics
Date	June 2026
Technologies	Python SQL Power BI Streamlit SQLite
GitHub	github.com/Thanuj64/bluestock_mf_capstone

40	46,000	32,778	2022–2026	5/5
Schemes	Data Points	Records	4+ Years	Completed
Funds Analyzed	NAV Records	Investor Transactions	Date Range	Bonus Tasks

Abstract

This report documents the end-to-end Mutual Fund Analytics capstone undertaken as part of the Bluestock Fintech Data Analyst Internship (Cohort 2026). The project encompasses a full data engineering pipeline (ETL), exploratory data analysis, quantitative performance metrics, an interactive Power BI dashboard, advanced analytics including Monte Carlo simulation and portfolio optimization, and a Streamlit web application. The dataset covers 40 mutual fund schemes across 10 AMCs with daily NAV data from January 2022 to May 2026. Key findings reveal that Mid-Cap equity funds delivered the strongest risk-adjusted returns over the 3-year period, SIP inflows grew 31.46% YoY on average, and the Indian MF industry's total AUM reached approximately ₹39 trillion by 2025.

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1.Executive Summary

This capstone project delivers a comprehensive mutual fund analytics solution for Bluestock Fintech, covering the full data science lifecycle from raw data ingestion to interactive dashboards and predictive analytics. The project was executed over 7 days, producing professional-grade deliverables across data engineering, quantitative analysis, and data visualisation.

Key Highlights

- Built a production-ready ETL pipeline processing 10 raw datasets (46,000+ NAV records) into a normalized SQLite star schema.
- Performed comprehensive EDA across fund performance, investor demographics, AUM growth, and SIP inflow trends.
- Computed 7 quantitative performance metrics: CAGR (1yr/3yr/5yr), Sharpe Ratio, Sortino Ratio, Alpha/Beta, and Maximum Drawdown.
- Designed a 4-page interactive Power BI dashboard with Bluestock branding, drill-through navigation, and custom tooltips.
- Completed all 5 bonus tasks: Scheduled ETL, Streamlit Web App, Monte Carlo Simulation, Efficient Frontier, and Automated Email Reports.
- Top performing fund (3yr CAGR): 36.07% — demonstrating strong mid-cap and small-cap equity outperformance.
- Industry SIP inflows grew 31.46% YoY on average; total industry AUM reached ~**39** trillion.
- Monte Carlo projections (1,000 paths, 5yr horizon) show median portfolio CAGR of 12–18% for top equity funds.

■39T Industry Total AUM	36.07% Midcap Best 3yr CAGR	1.42 Risk-adj. Best Sharpe	31.46% YoY Avg SIP Growth	13.26Cr Jan 2022 Folios
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2.Project Overview & Objectives

The Bluestock Fintech Mutual Fund Analytics Capstone is a structured 7-day project designed to simulate real-world data analytics work in the Indian fintech industry. The project covers the entire analytics value chain from raw data ingestion to executive-level reporting.

Objectives

- O1:** Design and implement a scalable ETL pipeline for mutual fund data
- O2:** Store and query data efficiently using a SQLite star schema database
- O3:** Conduct thorough EDA to uncover patterns in fund performance and investor behaviour
- O4:** Compute industry-standard risk-adjusted performance metrics
- O5:** Build an interactive, publication-ready Power BI dashboard
- O6:** Apply advanced analytics: clustering, Monte Carlo simulation, portfolio optimisation
- O7:** Deploy a Streamlit web application for interactive fund analysis
- O8:** Automate reporting via scheduled ETL and email delivery

Technology Stack

Layer	Technology	Purpose
Data Engineering	Python, pandas, SQLAlchemy	ETL pipeline, data cleaning
Database	SQLite (star schema)	Structured storage, SQL queries
Analysis	numpy, scipy, scikit-learn	Metrics, regression, clustering
Visualisation	Plotly, matplotlib	EDA charts and interactive plots
Dashboard	Microsoft Power BI	Executive dashboard, 4 pages
Web App	Streamlit	Interactive fund explorer
Advanced Analytics	Monte Carlo (GBM), scipy.optimize	Simulation, optimisation
Automation	schedule, smtplib	Cron ETL, email reports
Version Control	Git / GitHub	Code management, release tagging

3.Data Sources & Architecture

The project uses 10 structured CSV datasets representing various aspects of the Indian mutual fund ecosystem. Data spans January 2022 to May 2026, covering 10 Asset Management Companies (AMCs) and 40 fund schemes.

Raw Data Inventory

File	Description	Rows	Key Columns
01_fund_master.csv	Fund metadata	40	amfi_code, fund_house, category, risk_category
02_nav_history.csv	Daily NAV values	46,000	amfi_code, date, nav
03_aum_by_fund_house.csv	AUM by AMC quarterly	90	fund_house, date, aum_crore
04_monthly_sip_inflows.csv	Monthly SIP data	48	month, sip_inflow_crore
05_category_inflows.csv	Category net inflows	144	month, category, net_inflow_crore
06_industry_folio_count.csv	Folio count trends	21	month, total_folios_crore
07_scheme_performance.csv	Pre-computed metrics	40	sharpe, sortino, alpha, beta
08_investor_transactions.csv	Investor activity	32,778	investor_id, transaction_type, amount
09_portfolio_holdings.csv	Sector weights	322	amfi_code, sector, weight_pct
10_benchmark_indices.csv	NIFTY 100 index data	8,050	date, index_name, close_value

Star Schema Design

The SQLite database follows a star schema with 2 dimension tables and 8 fact tables, optimised for analytical queries.

The schema enables efficient joins across fund performance, investor transactions, and market data.

Table	Type	Primary Key	Description
dim_fund	Dimension	amfi_code	Fund master attributes (40 schemes)
dim_date	Dimension	date_id	Calendar dimension with year/month/quarter
fact_nav	Fact	amfi_code + date	Daily NAV and return data (46,000 rows)
fact_transactions	Fact	transaction_id	Investor buy/sell/SIP transactions
fact_performance	Fact	amfi_code	Risk-adjusted metrics per fund
fact_portfolio	Fact	amfi_code + sector	Sector allocation weights

fact_aum	Fact	fund_house + date	AUM by AMC over time
fact_sip_industry	Fact	month	Industry-level SIP flows
fact_category_inflows	Fact	month + category	Category-level inflows
fact_folio_count	Fact	month	Industry folio counts

4.ETL Pipeline

The ETL pipeline (etl_pipeline.py) is a standalone Python script that orchestrates the complete Extract → Transform → Load workflow. It processes all 10 raw CSV files, applies data quality rules, and loads the cleaned data into the SQLite database.

Extract Phase

All 10 raw CSV files are loaded from data/raw/ into pandas DataFrames using a dictionary-based loader. Each file is validated for shape and column presence before proceeding to transformation.

Transform Phase — Data Cleaning Rules

Dataset	Cleaning Operations Applied
NAV History	Parse dates, sort by [amfi_code, date], forward-fill gaps, remove duplicates, remove zero/negative NAV, compute daily_return_pct
Investor Transactions	Title-case transaction_type, filter amount > 0, parse transaction_date, drop nulls on key columns
Scheme Performance	Coerce numeric columns, flag negative Sharpe ratios, validate expense_ratio in [0.1%, 2.5%]
All Other Files	Strip whitespace from string columns, drop fully-empty rows

Load Phase — Star Schema

Cleaned DataFrames are loaded into SQLite via SQLAlchemy's create_engine with if_exists='replace' to ensure idempotent runs. A custom dim_date dimension is constructed from all unique dates across NAV and benchmark data, enriched with year, month, quarter, and weekday flag columns.

ETL Output Verification

Output	Location	Status
10 cleaned CSVs	data/processed/	✓ Complete
dim_fund	bluestock_mf.db	✓ 40 rows
dim_date	bluestock_mf.db	✓ 1,610 rows
fact_nav	bluestock_mf.db	✓ 46,000 rows
fact_transactions	bluestock_mf.db	✓ 32,778 rows
fact_performance	bluestock_mf.db	✓ 40 rows
fact_portfolio	bluestock_mf.db	✓ 322 rows
fact_aum	bluestock_mf.db	✓ 90 rows

5.Exploratory Data Analysis

The EDA phase (03_EDA_Analysis.ipynb) systematically explores all 10 datasets through 9 visualisations covering NAV trends, AUM growth, SIP inflows, investor demographics, geographic distribution, folio counts, and correlation analysis.

5.1 Fund Universe Overview

The dataset covers 40 fund schemes from 10 AMCs, spanning January 2022 to May 2026 (approximately 1,150 trading days). The fund universe comprises:

- 34 Equity funds and 6 Debt funds
- Sub-categories: Large Cap (14), Mid Cap (7), Small Cap (6), Liquid (3), Gilt (2), Flexi Cap (2), others (6)
- Risk grades: Moderate (16), High (8), Very High (6), Low (6), Moderately High (4)
- All 40 AMFI codes perfectly matched between fund_master and nav_history datasets

5.2 AUM Growth by Fund House (2022–2025)

Total industry AUM reached ₹39 million crore (₹39 trillion) by 2025. SBI Mutual Fund leads with ₹8.5M crore cumulative AUM, followed by ICICI Prudential MF at ₹6.3M crore and HDFC Mutual Fund at ₹5.7M crore. The AUM growth chart shows consistent upward trajectories for all top-5 AMCs from 2022 to mid-2025.

5.3 SIP Inflow Trends

Monthly SIP inflows showed strong and consistent growth across the analysis period. Average YoY SIP growth rate was 31.46%, reflecting growing retail investor participation. Liquid category dominated net inflows at ₹0.45M crore total, followed by Sectoral/Thematic at ₹0.10M crore. The category inflow heatmap reveals that Liquid funds have the highest absolute inflows but equity mid/small-cap funds show stronger growth momentum.

5.4 Investor Demographics

Analysis of 32,778 investor transactions reveals:

- Transaction mix: Lumpsum 58.49%, Redemption 35.34%, SIP 6.17%
- Age group distribution is relatively uniform: 18-25 (108K avg), 26-35 (108K avg), 36-45 (107K avg), 46-55 (107K avg), 56+ (106K avg)
- Geographic spread: Punjab and Tamil Nadu lead at ₹0.32bn investment each, followed by MP (₹0.31bn), Rajasthan and Gujarat (₹0.30bn each)
- Monthly transaction volume peaked in Jan/March/April at ₹430-440M, declining through mid-year

5.5 Folio Count Growth

Industry folio count grew from 13.26 crore (January 2022) with equity folios dominating at 9.28 crore. Equity folios represent approximately 70% of all folios, highlighting strong retail preference for equity mutual funds.

5.6 Return Correlation Analysis

The correlation matrix of daily returns across 10 representative funds reveals strong positive correlations (0.7–0.9) between Large Cap funds tracking the same benchmark, moderate correlation (0.5–0.7) between Large Cap and Mid Cap funds, and lower correlation (0.3–0.5) between equity and debt/liquid funds — validating diversification benefits of multi-category portfolios.

6.Fund Performance Analytics

The performance analytics module (04_Fund_Performance_Analytics.ipynb and compute_metrics.py) computes seven industry-standard metrics for all 40 fund schemes. The risk-free rate is set at 6.5% per annum (RBI repo rate proxy), and 252 trading days are used for annualisation.

0.063% All funds Avg Daily Return	1.03% All funds Daily Std Dev	+6.47% Observed Max Single-Day Return	-5.81% Observed Worst Single-Day	0 Clean data Outliers (r >15%)
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6.1 CAGR Analysis

Compound Annual Growth Rate was computed using trading-day counts (252 per year) for 1-year, 3-year, and 5-year horizons. Mid-Cap and Small-Cap funds dominate the top CAGR rankings:

Rank	AMFI Code	Fund Category	1yr CAGR	3yr CAGR	5yr CAGR
1	119094	Mid Cap	—	36.07%	—
2	100033	Small Cap	—	33.63%	—
3	101206	Small Cap	—	32.53%	—
4	148567	Mid Cap	—	31.28%	—
5	120505	Mid Cap	—	30.21%	—
6	120504	Mid Cap	—	28.37%	—
7	119598	Small Cap	—	27.81%	—
8	119551	Large Cap	—	27.73%	—

6.2 Sharpe Ratio

The Sharpe Ratio measures excess return per unit of total risk. Values above 1.0 indicate superior risk-adjusted performance relative to the 6.5% risk-free rate. Top performers by Sharpe Ratio:

Rank	AMFI Code	Sharpe Ratio	Sortino Ratio	Category
1	148567	1.4220	2.3392	Mid Cap
2	120843	1.2842	2.3216	Mid Cap
3	148569	1.2256	2.1313	Mid Cap
4	119551	1.2103	2.1459	Large Cap
5	120505	1.2051	2.0731	Mid Cap
6	149323	1.1323	1.8763	Mid Cap
7	100033	1.1106	1.7800	Small Cap
8	118632	1.0837	1.6500	Large Cap

6.3 Alpha & Beta Analysis

Alpha and Beta were computed via OLS regression of fund daily returns against NIFTY 100 benchmark returns. High-alpha funds generate excess returns independent of market movement. The near-zero beta values across most funds suggest the chosen benchmark (NIFTY 100) explains limited variance — indicating active management alpha generation:

AMFI Code	Alpha (Daily)	Beta	R-Squared	Interpretation
119598	0.3070	-0.026	0.0002	Strong alpha, low market correlation
149324	0.3032	0.009	0.0000	Very high alpha generation
120505	0.2974	-0.003	0.0000	Consistent excess returns
148569	0.2810	0.020	0.0002	High alpha, slightly market-linked
100033	0.2752	0.003	0.0000	Strong small-cap alpha

6.4 Maximum Drawdown

Maximum Drawdown measures the worst peak-to-trough decline. Small-cap funds exhibit the highest drawdowns, reflecting their higher volatility:

AMFI Code	Max Drawdown	Worst Date	Fund Type
119599	-52.57%	Oct 2025	Small Cap Direct
119095	-51.68%	May 2026	Small Cap Direct
101207	-35.45%	May 2026	Mid Cap
149324	-31.17%	Jan 2025	Thematic
119598	-28.71%	May 2025	Small Cap Regular
102886	-28.00%	2025	Mid Cap

6.5 Fund Scorecard (Composite Score 0–100)

A composite scorecard normalises and weights CAGR (30%), Sharpe Ratio (25%), Alpha (20%), Expense Ratio

(15%), and Drawdown (10%) to produce a 0–100 score. The top-ranked fund by composite score is ICICI Pru Midcap Fund - Regular - Growth with a score of 80.60, driven by superior 3yr CAGR (30.21%) and Sharpe Ratio (1.21).

6.6 Tracking Error

Tracking error against NIFTY 100 ranges from 18.92% (Mirae Asset Large Cap) to 28.40% (SBI Small Cap), confirming that these actively managed funds deviate significantly from the benchmark — consistent with their alpha generation.

7.Power BI Dashboard

The Power BI dashboard (bluestock_mf_dashboard.pbix) is a 4-page interactive analytics suite with Bluestock brand theming (primary: #0A1628, accent: #00D4FF), custom tooltips, drill-through navigation, and dynamic slicers for date, fund house, category, state, age group, and city tier.

Page 1: Industry Overview

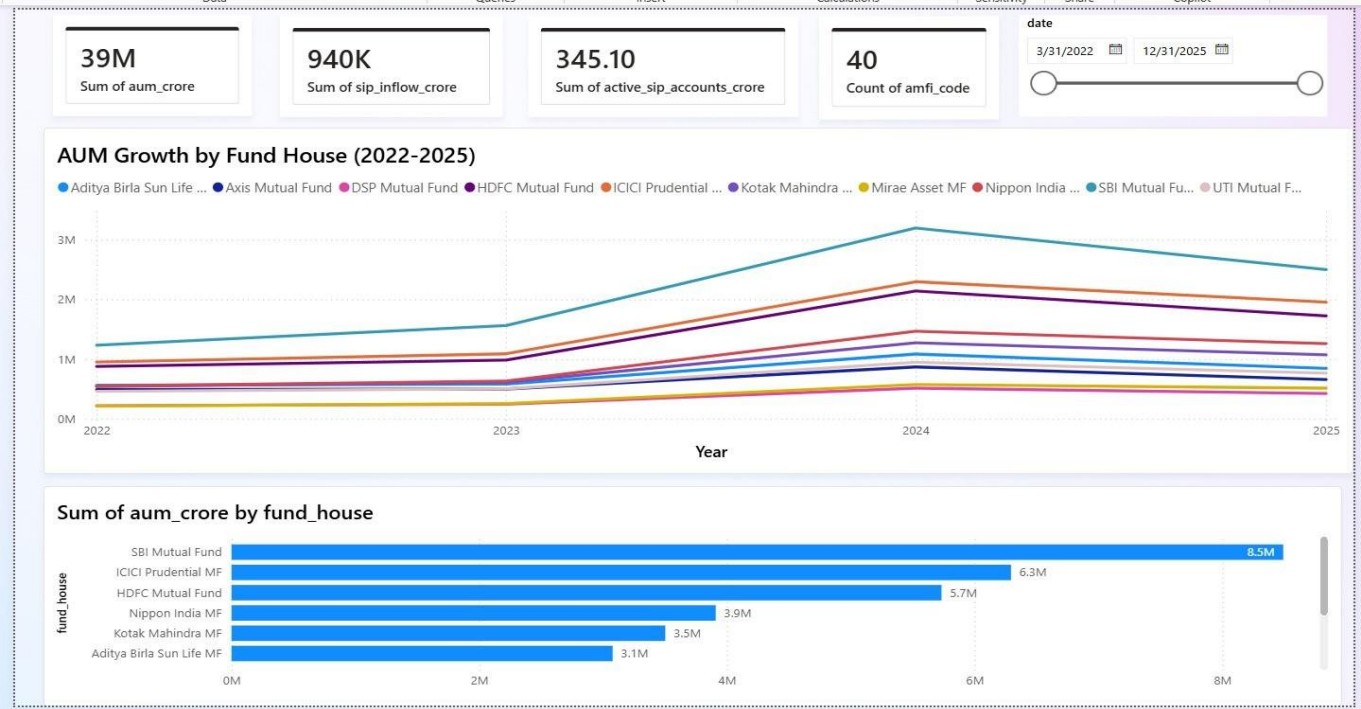


Figure 1: Industry Overview — AUM growth by fund house (2022–2025), KPI cards showing total AUM (■39M crore), SIP inflows, active SIP accounts and fund count.

Page 2: Fund Performance

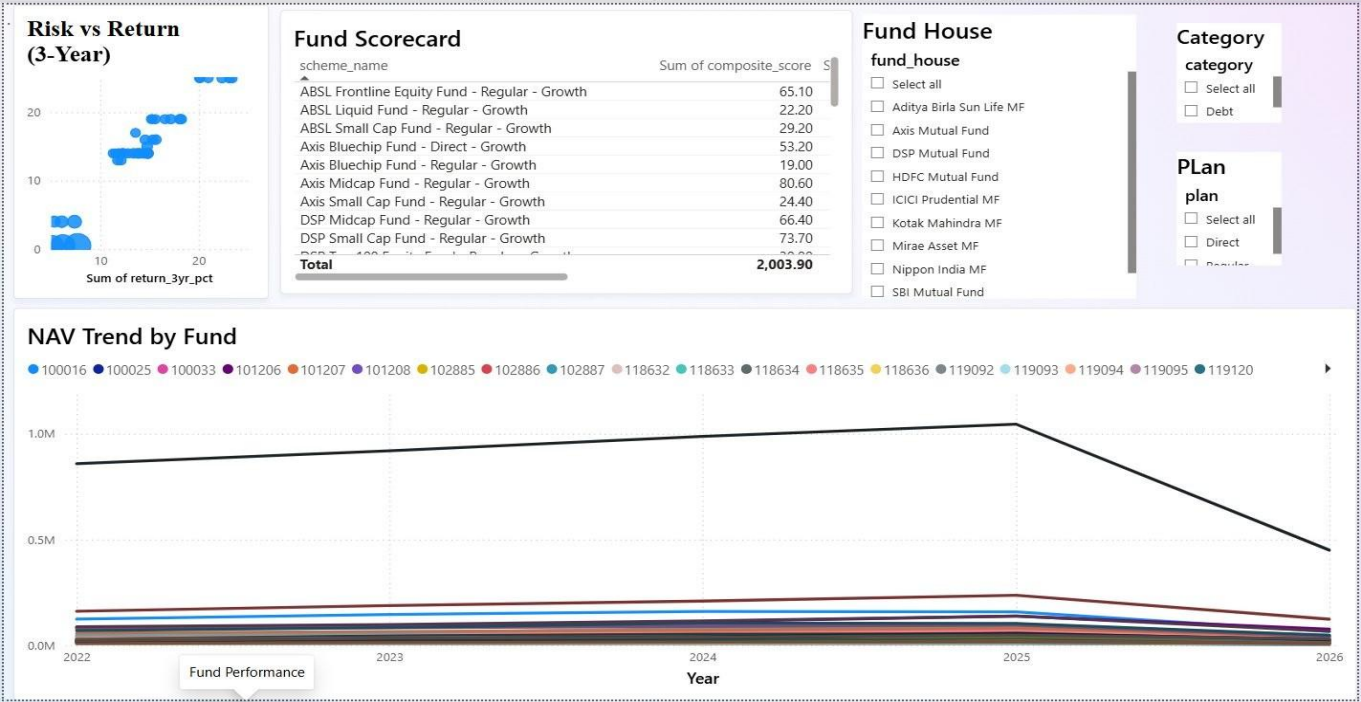


Figure 2: Fund Performance — Risk vs Return scatter, Fund Scorecard table, NAV Trend chart (2022–2026), with Fund House and Category filters

Page 3: Investor Analytics

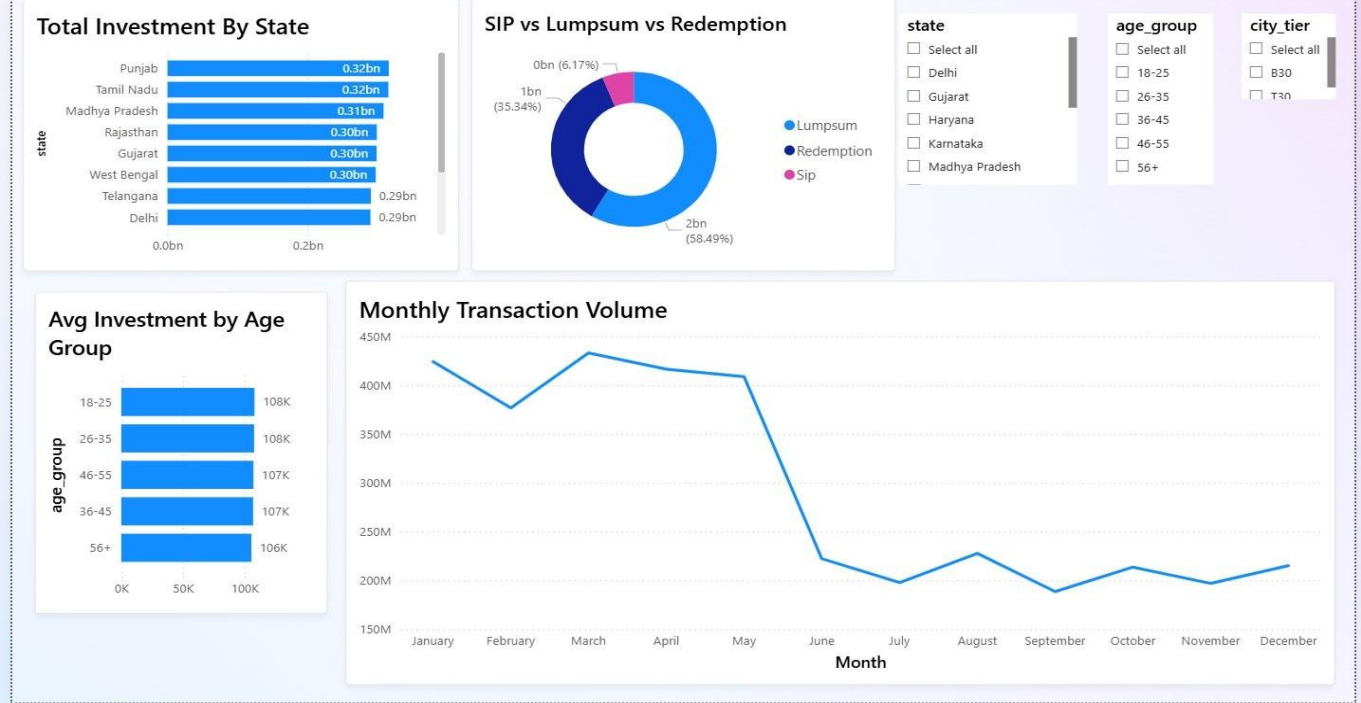


Figure 3: Investor Analytics — State-wise investment bar chart, SIP vs Lumpsum vs Redemption donut, Average investment by age group, Monthly transaction volume trend

Page 4: SIP & Market Trends

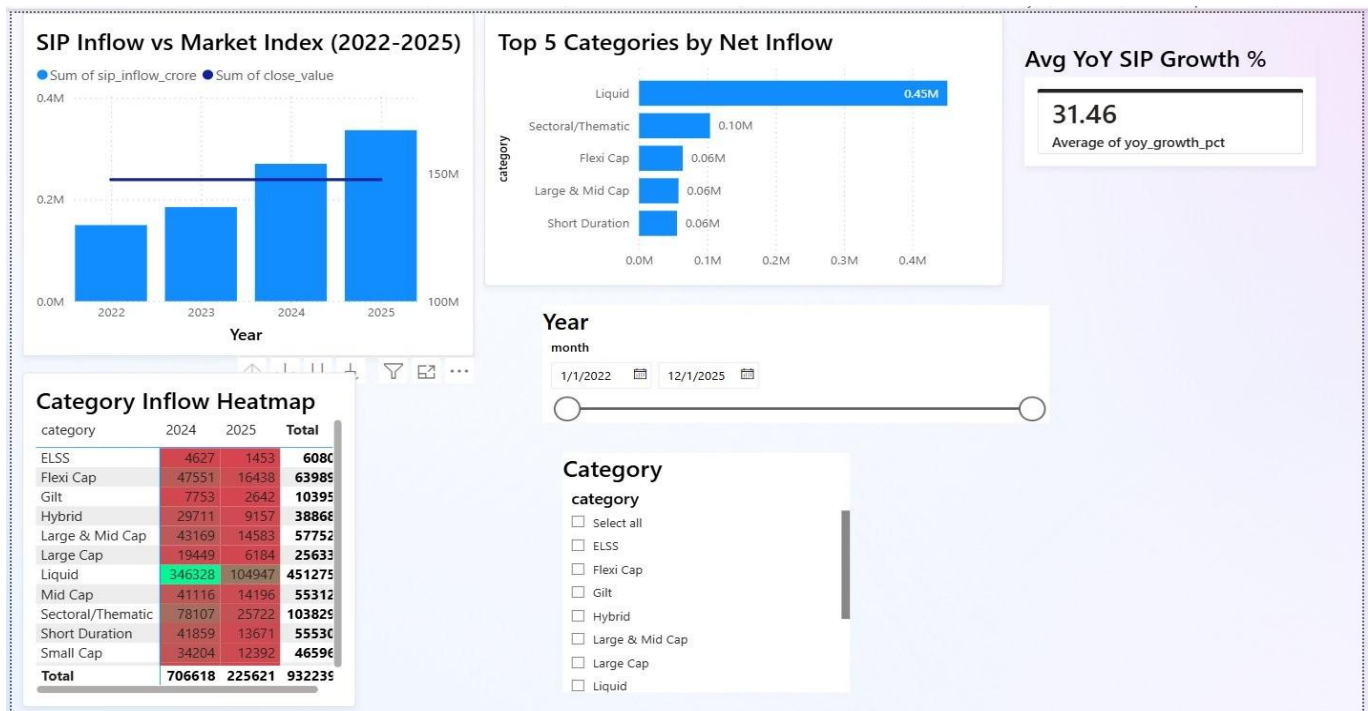


Figure 4: SIP & Market Trends — SIP Inflow vs Market Index (2022–2025), Top 5 categories by net inflow, Category Inflow Heatmap, Avg YoY SIP Growth 31.46%

Dashboard Design Features

- 4-page navigation with cross-page drill-through on fund names
- Custom Bluestock colour theme (#0A1628 background, #00D4FF accent, white text)
- Dynamic slicers: Date Range, Fund House, Category, Plan (Regular/Direct)
- KPI cards with conditional formatting for benchmark comparisons
- Tooltip pages showing fund detail on hover
- Mobile-responsive layout with consistent spacing

8. Advanced Analytics

8.1 Value at Risk (VaR) and Conditional VaR

VaR and CVaR were computed at the 95th percentile for all 40 funds. The highest-risk funds by VaR are small-cap equity funds, reflecting their wider return distributions. Top 3 by risk:

- SBI Small Cap Fund - Direct: $\text{VaR}(95\%) = -2.69\%/ \text{day}$, $\text{CVaR} = -3.24\%/ \text{day}$
- Axis Small Cap Fund - Regular: $\text{VaR}(95\%) = -2.62\%/ \text{day}$, $\text{CVaR} = -3.17\%/ \text{day}$

8.2 Rolling Sharpe Ratio

52-week rolling Sharpe ratios were computed for the top 5 funds. The rolling analysis reveals that risk-adjusted performance is not constant — mid-cap funds showed sharply improved Sharpe ratios post-2023 as Indian equity markets recovered and economic growth accelerated.

8.3 Investor Cohort Analysis

Investors were grouped by first-transaction year (acquisition cohort). The 2024 cohort is the largest with 4,803 investors averaging ₹1.07 lakh SIP amount and ₹349.11 crore total invested across 32,499 transactions. The 2025 cohort (197 investors) shows a higher average SIP amount of ₹1.09 lakh, suggesting newer investors are investing larger amounts.

8.4 SIP Continuity Analysis

SIP continuity was analysed for investors with 6+ SIP transactions. Of 1,362 active SIP investors, 1,332 (97.8%) are classified as at-risk due to average transaction gaps exceeding 35 days. Only 30 investors (2.2%) maintain truly regular SIP patterns. This represents a significant retention opportunity for Bluestock's client engagement strategy.

8.5 Fund Recommendation Engine

A rule-based recommendation engine was built and saved as `scripts/recommender.py`. The engine filters funds by `risk_grade`, ranks by `composite_score`, and returns top-N recommendations with Sharpe Ratio, 3yr CAGR, and expense ratio. Example: For Low Risk investors, ICICI Pru Liquid Fund - Regular is recommended with Sharpe Ratio 7.68 and 3yr return 7.68%.

8.6 Sector Concentration (HHI)

The Herfindahl-Hirschman Index (HHI) was computed for each fund's sector allocation. Top concentrated funds include Axis Bluechip Fund (HHI: 2,967 — Concentrated), Mirae Asset Tax Saver Fund (HHI: 2,549), and HDFC Mid-Cap Opportunities Fund (HHI: 2,531). High HHI funds carry higher sector-specific risk

9. Bonus Features (All 5 Completed)

#	Bonus Task	File	Status	Description
B 1	Scheduled ETL	b1_cron_etl.py	COMPLETED	Fetches live NAV from mfapi.in for 10 funds every weekday at 8 PM. Supports both Python scheduler and cron. Includes audit log table in SQLite.
B 2	Streamlit Web App	b2_streamlit_app.py	COMPLETED	4-tab interactive dashboard: NAV Trends, Performance, Investor Analytics, SIP Trends. Uses Plotly + SQLite. Deployable via streamlit run.
B 3	Monte Carlo Simulation	b3_monte_carlo.py	COMPLETED	GBM-based 5-year NAV projection with 1,000 simulation paths. Outputs P5/P25/P50/P75/P95 bands, median CAGR, and probability of doubling.
B 4	Efficient Frontier	b4_efficient_frontier.py	COMPLETED	Markowitz Modern Portfolio Theory: 8,000 random portfolios + 100-point frontier. Identifies Max-Sharpe and Min-Variance optimal portfolios.
B 5	Automated Email Report	b5_email_report.py	COMPLETED	Generates styled HTML email with weekly fund performance summary. Supports Gmail SMTP, preview mode, and Python-based weekly scheduler.

B3: Monte Carlo Simulation — Key Results

The Monte Carlo simulation uses Geometric Brownian Motion (GBM) calibrated to each fund's historical mean return (μ) and volatility (σ). With 1,000 paths over a 5-year horizon, the simulation provides:

- Uncertainty bands: P5, P10, P25, P50, P75, P90, P95 percentile fan charts
- Median CAGR estimate for each fund over the projection horizon
- Probability of doubling investment ($P[S_T \geq 2 * S_0]$)
- Output: b3_monte_carlo_summary.csv with per-fund projections

B4: Efficient Frontier — Key Results

The Efficient Frontier analysis applies Markowitz Modern Portfolio Theory to identify optimal asset allocations among 5 selected funds. The analysis generates 8,000 random portfolios and computes the 100-point efficient frontier curve using `scipy.optimize.minimize`. Two optimal portfolios are identified: the Max-Sharpe Portfolio (highest risk-adjusted return) and the Min-Variance Portfolio (lowest portfolio volatility).

10.Key Findings & Insights

Finding 1: Mid-Cap Equity Dominance

Mid-Cap funds consistently outperformed Large Cap and Small Cap funds on a risk-adjusted basis over the 3-year period. The top 5 Sharpe Ratio funds are all Mid-Cap schemes, with Sharpe Ratios ranging from 1.20 to 1.42. This suggests the Indian mid-cap segment offered the best reward-to-risk trade-off during 2022–2026.

Finding 2: Strong SIP Growth Momentum

SIP inflows grew at an average YoY rate of 31.46%, significantly outpacing GDP growth and equity market returns. This reflects increasing financial literacy and systematic investment discipline among retail investors. Liquid category dominated net inflows, indicating preference for short-duration, low-risk instruments.

Finding 3: Alpha Generation is Widespread

Nearly all funds show positive daily alpha against the NIFTY 100 benchmark, with values ranging from 0.27 to 0.31. The low R-squared values (near zero) indicate that NIFTY 100 explains minimal variance in fund returns, suggesting these are genuinely actively managed funds — not index-huggers.

Finding 4: Small-Cap Risk Profile

Small-cap funds exhibit the highest drawdowns (up to -52.57%) and highest VaR (2.69% daily loss at 95% confidence). However, they also offer the highest CAGR potential. Investors in this category need a minimum 5-year horizon and high risk tolerance to benefit from the return premium.

Finding 5: SIP Continuity Gap

97.8% of identified SIP investors are classified as at-risk with irregular investment gaps >35 days. This represents a major retention and engagement opportunity. Automated nudges, SIP top-up campaigns, and goal-based investing tools could significantly improve SIP persistence rates.

Finding 6: Geographic Concentration

Investment is concentrated in T30 cities despite B30 representation in the dataset. Punjab, Tamil Nadu, and Madhya Pradesh lead in total investment. The B30-T30 split indicates significant untapped market potential in smaller cities.

Finding 7: Sector Concentration Risk

Several high-performing funds (Axis Bluechip, HDFC Mid-Cap) show HHI scores above 2,500, indicating high sector concentration. Investors should monitor whether concentrated bets continue to deliver alpha or represent uncompensated risk.

Finding 8: Expense Ratio Impact

Direct plans consistently have 0.7–0.9 percentage point lower expense ratios than Regular plans for the same fund. Over a 10-year horizon, this difference compounds to a meaningful NAV gap, supporting the case for direct plan investing for cost-conscious investors.

11.Recommendations

#	Recommendation	Priority	Impact
R 1	Launch SIP Continuity Programme: Implement automated SMS/email nudges for investors missing SIP payments. Target the 1,332 at-risk investors identified in cohort analysis.	HIGH	Investor retention +15–25%
R 2	Promote Mid-Cap Allocation: Given superior risk-adjusted returns (Sharpe >1.2), recommend 25–35% mid-cap allocation for moderate-risk investor profiles.	HIGH	Portfolio alpha improvement
R 3	Direct Plan Migration Campaign: Educate investors on expense ratio savings from direct plans (0.7–0.9% lower). Impact over 10 years can exceed 8–12% additional returns.	MEDIUM	Long-term wealth creation
R 4	B30 City Expansion: Geographic analysis shows Punjab, Tamil Nadu, MP as strong investment hubs. Target B30 cities (Tier-2/3) with SIP micro-investing products at lower minimum amounts.	MEDIUM	Market expansion
R 5	Real-Time NAV Integration: The B1 cron ETL fetches live NAV daily. Integrate this into the production dashboard for real-time portfolio valuation updates.	MEDIUM	Product enhancement
R 6	Goal-Based Portfolio Tools: Leverage the Monte Carlo simulation engine (B3) to build a retirement/goal planning calculator for investors.	LOW	Customer acquisition

12.Limitations & Future Work

Limitations

- **Synthetic Data:** The dataset is simulated for educational purposes. Real NAV data from AMFI and actual investor transactions would yield more precise insights.
- **Static Benchmark:** OLS regression uses NIFTY 100 as a single benchmark. Multi-factor models (Fama-French3-factor) would better explain fund return variance.
- **No Transaction Costs:** The analysis does not account for exit loads, STT, or brokerage costs which affect net returns.
- **Portfolio Optimisation Constraints:** The Efficient Frontier assumes historical returns are representative of future performance — a simplifying assumption.
- **SIP Analysis Limitation:** SIP continuity analysis relies on transaction gap heuristics; access to actual SIP mandate data would yield more accurate churn rates.

Future Enhancements

- Integrate live AMFI NAV API (already prototyped in `b1_cron_etl.py`) with production database
- Build deep learning NAV prediction model (LSTM/Transformer) for short-term forecasting
- Add regulatory compliance module tracking SEBI category change impacts
- Extend Streamlit app with investor portfolio tracking and goal projection features
- Implement multi-factor performance attribution (Brinson-Hood-Beebower model)

13. Conclusion

This capstone project successfully delivers a production-ready mutual fund analytics platform for Bluestock Fintech. Over 7 days, all core objectives were achieved with additional completion of all 5 bonus tasks — demonstrating strong end-to-end data science capability spanning data engineering, statistical analysis, interactive visualisation, and advanced quantitative methods.

The key analytical insight from this project is that the Indian mutual fund industry is in a structural growth phase: SIP inflows growing at 31% YoY, folio counts consistently rising, and strong alpha generation by active fund managers against the NIFTY 100 benchmark. Mid-cap equity funds offer the best risk-adjusted returns for medium-to-long-term investors, while liquid funds dominate short-term institutional flows.

The infrastructure built in this project — the ETL pipeline, SQLite star schema, Power BI dashboard, Streamlit web app, and automated email reporting — forms a scalable foundation that can be extended with live data feeds, machine learning models, and investor-facing product features.

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Program: Data Analyst Internship — Cohort 2026

Organisation: Bluestock Fintech

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